

A Two-Stage Perceptual Account of Interval Consonance

David De Roure

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Abstract

Euler's *Gradus Suavitatis* (1739) assigns a dissonance value to a musical interval p/q by the formula $G(p/q) = 1 + \Omega^*(p) + \Omega^*(q)$, where $\Omega^*(n) = \sum e_i(p_i - 1)$ sums the weighted prime exponents of n . We show that the simpler asymmetric formula $f(p/q) = p + \Omega^*(q)$ achieves a higher rank correlation with human consonance ratings ($\rho = 0.989$ vs 0.979), and that this formula admits a natural two-stage perceptual interpretation grounded in harmonic series theory. We also show that *Gradus* is exactly equivalent to a weighted harmonic coincidence count with weights equal to $\Omega^*(n)$ itself, and that the simpler metric $\max(p, q)$ — equivalent to Galileo's pulse-coincidence model — achieves the same correlation as $f(p/q)$.

1. Background

Euler's *Gradus Suavitatis*, catalogued in OEIS A275314, assigns to each musical interval a positive integer measuring its arithmetic dissonance. For a ratio p/q in lowest terms, the characteristic number is $n = p \cdot q$, and:

$$G(p/q) = 1 + \Omega^*(p \cdot q)$$

where Ω^* is the completely additive function $\Omega^*(n) = \sum e_i(p_i - 1)$. Since Ω^* is completely additive, $G(p/q) = 1 + \Omega^*(p) + \Omega^*(q)$, treating numerator and denominator symmetrically. The weights $(p - 1)$ for prime p encode Euler's aesthetic judgment that large primes represent greater “difficulty.”

Euler's formula correctly ranks the 13 standard intervals of Western music against human consonance ratings (Krumhansl 1990) with Spearman $\rho = 0.979$, and is exactly tied only on the Major third / Major sixth pair and on a three-way tie at *Gradus* 8.

2. *Gradus* as Harmonic Coincidence

For two notes at frequency ratio $p:q$ (coprime), exact harmonic coincidences occur at harmonics $p, 2p, 3p, \dots$ of the lower note. For a fixed count M of coincidences, the weighted sum with weights $w(n) = \Omega^*(n)$ yields:

$$\sum_{k=1}^M \Omega^*(k \cdot p) = \Omega^*(p) \cdot M + C(M)$$

where $C(M) = \sum_{k=1}^M \Omega^*(k)$ is a constant independent of p . This follows from the complete additivity of Ω^* : $\Omega^*(k \cdot p) = \Omega^*(k) + \Omega^*(p)$. The symmetric score over both notes gives:

$$\text{Score} = M \cdot (G(p/q) - 1) + 2 \cdot C(M)$$

***Gradus* is therefore exactly a weighted harmonic coincidence count**, with harmonics weighted by their own prime complexity. This confirms the physical intuition behind Euler's formula but

reveals its self-referential structure: recovering Gradus from coincidences requires weights that are themselves Gradus-like.

Unweighted coincidence counting (flat weights) achieves $\rho = 0.791$ across all coprime ratios with $p, q \leq 16$. The weight function $\log(n)$ achieves $\rho = 0.795$ without any prime structure, establishing a ceiling for physics-based models that do not invoke prime factorisation.

3. Galileo's Model and $\max(p, q)$

Galileo (1638) described consonance in terms of pulse coincidences: two strings at ratio $p:q$ produce coinciding pulses every p vibrations of the lower string. The "sweetness" of the interval grows with the frequency of coincidences, i.e. inversely with p . The metric $\max(p, q) = p$ is therefore:

1. **Galileo's pulse model** — vibrations between coincidences
2. **Harmonic series position** — the upper note is partial p of the implied fundamental (Rameau 1722)
3. **Farey index** — the smallest n such that q/p appears in the Farey sequence F_n (Cauchy 1816)
4. **First coincidence harmonic** — the position in the lower note's series where the two notes first meet

These four descriptions, discovered independently across four centuries, are mathematically identical. $\max(p, q)$ achieves $\rho = 0.989$ against human ratings, marginally exceeding Gradus, but cannot distinguish intervals with the same numerator (e.g. $5/4$ and $5/3$ both have $\max = 5$).

4. An Improved Formula

Euler's formula is symmetric: $\Omega^*(p) + \Omega^*(q)$. Human ratings, however, distinguish numerator and denominator — the bass note and the upper note play different perceptual roles. We find empirically that the asymmetric formula:

$$f(p/q) = p + \Omega^*(q)$$

achieves $\rho = 0.989$, matching $\max(p, q)$ and exceeding Gradus. The formula differs from Gradus only in replacing $\Omega^*(p)$ with p for the numerator. Optimising prime weights freely with six parameters (separate weights for primes 2, 3, 5 in numerator and denominator) achieves perfect rank correlation $\rho = 1.000$, confirming that the structure required to perfectly predict the human rankings is present in an asymmetric prime-weighted formula.

The formula $f(p/q) = p + \Omega^*(q)$ can be derived from first principles as the sum of two quantities:

Component	Formula	Meaning
Upper note reach	p	Partial number of upper note in implied harmonic series
Bass context depth	$\Omega^*(q)$	Prime complexity of lower note's position in series

5. A Two-Stage Perceptual Model

The formula motivates a two-stage model of how the auditory system evaluates an interval:

Stage 1 — Bass establishes harmonic context (cost $\Omega^*(q)$).

The lower note is the q -th partial of an implied fundamental F . The brain extrapolates downward to F by “dividing out” the prime factors of q . Each factor of 2 (an octave) costs 1 unit; each factor of 3 costs 2; each factor of 5 costs 4. This is exactly $\Omega^*(q)$. Intervals whose bass note is a power of 2 (e.g. the octave, $q=1$; the major second bass, $q=8=2^3$) establish the fundamental cheaply via octave equivalence. Intervals with an odd prime in q (the fourth, $q=3$; the minor third bass, $q=5$) require the brain to extrapolate through a genuinely new prime, incurring higher cost.

Stage 2 — Upper note reaches into the series (cost p).

Once the fundamental is established, the upper note must be recognised as its p -th partial. Higher partials are quieter (amplitude proportional to $1/p$ in typical harmonic sounds), more narrowly tuned, more sensitive to mistuning, and have lower pitch salience (Terhardt 1979). The cost of confirming the upper note as a member of the harmonic series is therefore p itself — the direct measure of its height in the series.

Total perceptual effort: $f(p/q) = p + \Omega^*(q)$.

This account connects: - **Galileo** (1638): pulse coincidences as physical mechanism - **Rameau** (1722): the *corps sonore* (harmonic series) as the source of musical meaning - **Euler** (1739): prime arithmetic as the language of harmonic complexity - **Terhardt** (1979): virtual pitch and harmonic template matching as auditory mechanisms - **Huron** (2001): statistical learning of harmonic patterns as the basis of consonance judgments

6. The Remaining Failure

No simple formula distinguishes the Major third ($5/4$, human rank 5) from the Major sixth ($5/3$, human rank 7). Both have $p = 5$ (upper note is $P5$) and $\Omega^*(q) = 2$ (denominator is either $4 = 2^2$ or 3 , both with $\Omega^* = 2$). The distinction requires knowing that $q = 4 = 2^2$ (a pure octave stack, “free” under octave equivalence) is simpler than $q = 3$ (the first genuinely new prime), yet $\Omega^*(4) = \Omega^*(3) = 2$ so Euler’s weighting cannot see this. Resolving this tie requires either a richer formula or an appeal to learned cultural context (thirds occur far more frequently than sixths in tonal music).

7. Summary

Formula	q vs human	Notes
Euler’s Gradus: $1 + \Omega^*(p) + \Omega^*(q)$	0.979	Symmetric; two ties unresolved
Galileo / $\max(p,q)$	0.989	Simplest; same tie
$f(p/q) = p + \Omega^*(q)$	0.989	Asymmetric; two-stage interpretation
Optimised asymmetric weights	1.000	Six parameters; no closed form

Formula	q vs human	Notes
Roughness (Plomp-Levelt)	0.709–0.863	Register-dependent; fails on Tritone

The formula $f(p/q) = p + \Omega^*(q)$ is marginally simpler than Gradus, slightly more accurate, and directly interpretable as a two-stage process of harmonic extrapolation. It suggests that Euler’s prime-weighting is correct for the *denominator* but that the *numerator* is better described by the raw partial number — consistent with the auditory system’s known sensitivity to partial height rather than prime structure in the upper voice.

Scripts and data: /Users/daviddereure/gradus/